

PROJECT REPORT OF INDUSTRY ORIENTED HANDS ON EXPERIENCE (IOHE)

ON

**NutriGuide AI – AI Powered Nutrition and Meal Tracking Platform
submitted in partial fulfilment of the requirements for the award of degree of**

BACHELOR OF ENGINEERING

In

COMPUTER SCIENCE AND ENGINEERING

Submitted by:

Arayana Singh (2210990149)

Annanya (2210990130)

Bhavika Arora (2210990220)

Supervised By:

Dr. Gifty Gupta

Assistant Professor

CSE Dept.

Chitkara University, Punjab



DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

**CHITKARA UNIVERSITY INSTITUTE OF ENGINEERING AND TECHNOLOGY
CHITKARA UNIVERSITY, PUNJAB, INDIA**

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DECLARATION

I hereby certify that the work which is being presented in the project report entitled “**NutriGuide AI**” in partial fulfilment of requirement for the award of the degree of Bachelor of Engineering (Computer Science and Engineering) submitted in the department of Computer Science and Engineering at Chitkara University Institute of Engineering and Technology, Chitkara University, Punjab, India, is an authentic record of my own work carried out under the supervision of Dr. Gifty Gupta. The matter presented in this project report has not been submitted in any other university/institute for the award of any degree.

Place: Rajpura

Date:

Arayana Singh

2210990149

Annanya

2210990130

Bhavika Arora

2210990220

This is to certify that the above statement made by the candidate is correct to the best of my knowledge and belief.

Dr. Swati Goel

Assistant Professor

Department of Computer Science and Engineering

Chitkara University Institute of Engineering and Technology,

Chitkara University, Punjab, India

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Arayana Singh
(2210990149)

Annanya
(2210990130)

Bhavika Arora
(2210990220)

ABSTRACT

NutriGuide AI is an advanced and intelligent nutrition management system designed to promote healthy eating habits and simplify personalized diet planning through modern web technologies and Artificial Intelligence. Built using cutting-edge technologies such as React.js for a dynamic and responsive user interface, Node.js and Express.js for efficient backend operations, and MongoDB for secure and scalable data management, NutriGuide AI delivers a seamless and user-friendly experience across multiple devices. The platform is thoughtfully designed to assist users in tracking their nutrition, monitoring calorie intake, and receiving AI-powered diet recommendations based on their personal health goals and preferences.

Users benefit from a comprehensive set of features including secure registration, authentication, personalized profile management, meal tracking, calorie monitoring, and AI-generated nutrition guidance. The application goes beyond traditional calorie tracking systems by integrating Generative AI technology that provides intelligent meal recommendations, healthy eating suggestions, and customized diet plans. This AI-powered system analyses user preferences and nutritional requirements to generate meaningful and personalized health guidance, helping users make informed lifestyle decisions.

The platform also includes an interactive dashboard where users can monitor daily calorie consumption, nutrition progress, and meal history in an organized manner. Real-time API communication enables smooth interaction between frontend and backend modules, ensuring efficient data retrieval and updates. The responsive user interface is carefully designed to provide accessibility and smooth performance across desktops, tablets, and mobile devices.

NutriGuide AI demonstrates the practical implementation of modern full-stack web development and AI integration in the healthcare and fitness domain. By combining responsive design, secure authentication, intelligent recommendation systems, and real-time nutrition tracking, the platform provides an effective digital solution for maintaining healthier lifestyles and improving overall wellness.

1. Introduction

1.1. Background

In today's digital era, maintaining a healthy lifestyle has become increasingly challenging due to busy schedules, unhealthy eating habits, and lack of proper nutritional awareness. As people become more health-conscious, there is a growing demand for intelligent and personalized nutrition management systems that can provide real-time dietary guidance and calorie tracking. Modern users expect platforms that are not only visually appealing and easy to use, but also capable of delivering smart recommendations tailored to their individual health goals and lifestyle preferences.

Traditional diet management methods often rely on generic meal plans and manual calorie calculations, which can be time-consuming, inaccurate, and difficult to maintain consistently. Many existing fitness and nutrition applications provide limited personalization and lack intelligent features that adapt to user needs. NutriGuide AI addresses these limitations by offering an advanced AI-powered nutrition and meal tracking platform developed using modern technologies such as React.js, Node.js, Express.js, MongoDB, and Generative AI APIs. The application provides a responsive and interactive interface that enables users to manage their nutritional habits efficiently across multiple devices.

At the core of NutriGuide AI is an intelligent recommendation system that analyses user preferences, dietary goals, and nutritional requirements to generate personalized meal suggestions and healthy eating guidance. Users can securely register and log into the platform, track their calorie intake, maintain meal records, and interact with an AI-powered assistant that offers customized nutrition advice. The platform also features a dynamic dashboard where users can monitor their progress, daily calorie consumption, and dietary habits in an organized and visually appealing manner.

The application follows a full-stack architecture where the frontend is developed using React.js for building reusable and responsive UI components, while Node.js and Express.js manage backend APIs and server-side functionality. MongoDB is used for efficient and

scalable database management, ensuring secure storage of user information, meal records, and nutrition logs. Axios is integrated for seamless communication between frontend and backend services, enabling real-time updates and smooth user interaction.

NutriGuide AI is more than just a calorie tracking application—it is a smart digital nutrition assistant designed to promote healthier lifestyles through Artificial Intelligence and modern web technologies. The project demonstrates the practical implementation of full-stack development, responsive design principles, secure authentication systems, and AI integration in the healthcare and wellness domain. As technology continues to transform healthcare and fitness industries, NutriGuide AI provides a foundation for future innovations in personalized nutrition and intelligent health management systems.

1.2. Problem Statement

Despite the increasing popularity of fitness and nutrition applications, many existing platforms still fail to provide a truly personalized, intelligent, and user-friendly experience. Most traditional diet tracking systems focus only on basic calorie counting and static meal plans without understanding the unique nutritional requirements, health goals, and lifestyle preferences of individual users. As a result, users often struggle to maintain consistency in their diet routines and fail to receive meaningful guidance that supports long-term healthy habits.

One of the major challenges in existing nutrition management applications is the lack of intelligent recommendation systems capable of providing customized meal suggestions and nutrition guidance. Many applications provide generalized diet plans that may not be suitable for every individual. NutriGuide AI addresses this problem by integrating Generative AI technology that analyses user preferences and health-related inputs to generate personalized diet recommendations and healthy eating suggestions. This AI-driven approach improves user engagement and helps individuals make informed nutritional decisions based on their specific requirements.

Another significant issue with many existing applications is poor user experience and limited responsiveness across devices. Several platforms have outdated interfaces, complex navigation systems, and lack accessibility features necessary for smooth interaction. NutriGuide AI overcomes these challenges by implementing a modern responsive user interface using React.js and contemporary frontend design principles. The application ensures seamless performance across desktops, tablets, and mobile devices, providing users with an intuitive and interactive experience.

Additionally, many nutrition applications fail to maintain organized user data and secure authentication systems. NutriGuide AI uses MongoDB for scalable database management and JWT-based authentication for secure login and user session handling. The backend architecture developed using Node.js and Express.js ensures efficient API communication, secure data handling, and reliable application performance. This robust system architecture enables smooth interaction between frontend and backend modules while maintaining data consistency and scalability.

The project aims to bridge the gap between traditional calorie tracking applications and intelligent nutrition management systems by combining Artificial Intelligence, responsive design, secure authentication, and real-time data management into a single integrated platform. NutriGuide AI not only simplifies meal tracking and nutrition monitoring but also encourages users to adopt healthier lifestyles through personalized and intelligent digital assistance.

2. Methodology

2.1. Methods

The development methodology for NutriGuide AI follows a modular and component-based architecture using React.js, allowing the creation of reusable UI components and efficient state management throughout the application. The frontend interface is designed with responsive styling technologies such as Tailwind CSS and modern CSS principles to ensure a clean, interactive, and user-friendly experience across multiple devices. React components are structured in a scalable manner, enabling easier maintenance, debugging, and future feature expansion.

The backend functionality and API communication are handled using Node.js and Express.js, providing a fast and efficient server-side environment for processing user requests, authentication, and nutrition-related operations. MongoDB is used as the primary database system to securely store user profiles, meal records, calorie tracking information, and AI interaction data. Mongoose is integrated to simplify schema management and database operations while maintaining data consistency and scalability.

NutriGuide AI also incorporates Artificial Intelligence integration through Generative AI APIs to provide personalized nutrition guidance and intelligent meal recommendations. The AI system analyses user preferences, dietary goals, and nutrition-related inputs to generate customized suggestions that help users maintain healthier eating habits. Axios is used for smooth API communication between frontend and backend modules, enabling real-time data retrieval and updates without affecting user experience.

The application is developed iteratively with continuous testing and debugging throughout the development lifecycle. Both manual testing and functional testing techniques are used to verify authentication, API communication, database connectivity, and user interface responsiveness. This testing process helps identify and resolve errors efficiently while ensuring stable application performance.

The architecture of NutriGuide AI emphasizes maintainability, scalability, and modularity, allowing new features and functionalities to be integrated with minimal changes to the existing codebase. By separating the application into independent frontend and backend modules, developers can work on individual features such as authentication, meal tracking, AI recommendations, and dashboard management more effectively. State management and API handling are carefully optimized to ensure smooth real-time interaction and improved responsiveness throughout the platform.

Furthermore, responsive design principles are implemented to guarantee accessibility across desktops, tablets, and mobile devices. The project combines modern web technologies, secure authentication systems, AI integration, and scalable backend architecture to deliver a reliable and intelligent nutrition management solution.

3. Tools and Technologies Used

3.1. Programming / Working Environment

The application is developed using a modern full-stack JavaScript ecosystem that ensures scalability, performance, and ease of maintenance.

FRONTEND TECHNOLOGIES

Component	Specification
Frontend Framework	React.js (with JSX)
Styling	Tailwind CSS
API Communication	Axios
Build Tool	Vite
Icons	React Icons
Language	JavaScript

Package Manager	npm
Version Control	Git and GitHub
Development Tools	Visual Studio Code, Chrome DevTools
State Management	React Hooks

BACKEND AND DATABASE TECHNOLOGIES

Component	Specification
Backend Framework	Express.js and Node.js
Database	MongoDB
Database ODM	Mongoose
Authentication	JWT Authentication
AI Integration	Google Generative AI APIs
API Architecture	REST APIs
Environment Management	dotenv
Testing and Debugging	Postman

4. Database Analysing, design and implementation

In the development of NutriGuide AI, efficient database management plays a crucial role in delivering a personalized and intelligent nutrition management experience. Since the application continuously processes user information, meal records, calorie tracking data, and AI-generated nutrition recommendations, a robust and scalable database system is essential for ensuring smooth performance and secure data handling. MongoDB is used as the primary database solution because of its flexibility, scalability, and efficient handling of real-time data operations. The database architecture supports critical functionalities including user authentication, meal tracking, nutrition monitoring, AI interaction history, and profile management.

The database design of NutriGuide AI is centered around multiple collections dedicated to different functionalities within the system. The major collections include Users, Meals, Nutrition Logs, Calorie Records, and AI Chat History. Each user profile stores essential information such as personal details, health goals, dietary preferences, authentication credentials, and progress data. Meal records maintain information related to food intake, calorie values, nutritional details, and meal timing, allowing users to monitor their eating habits effectively. AI-generated recommendations and nutrition guidance are also stored to maintain interaction history and improve personalization.

The application workflow follows a data-driven approach where frontend operations interact with backend APIs, which further communicate with the MongoDB database for retrieving and storing information. Whenever a user logs in, tracks meals, updates profile data, or requests AI-based nutrition guidance, the backend server processes these requests and performs database operations efficiently. This ensures real-time responsiveness, smooth user interaction, and consistent data synchronization across the platform.

Mongoose is integrated as an Object Data Modeling (ODM) library for MongoDB to simplify schema creation, validation, and database management. Using Mongoose schemas, the project maintains structured data organization, secure validation rules, and consistency throughout the application. Schema-based data management also improves scalability and allows additional functionalities to be integrated easily in future development stages. JWT authentication mechanisms are connected with database operations to securely manage user login sessions and protect sensitive user information.

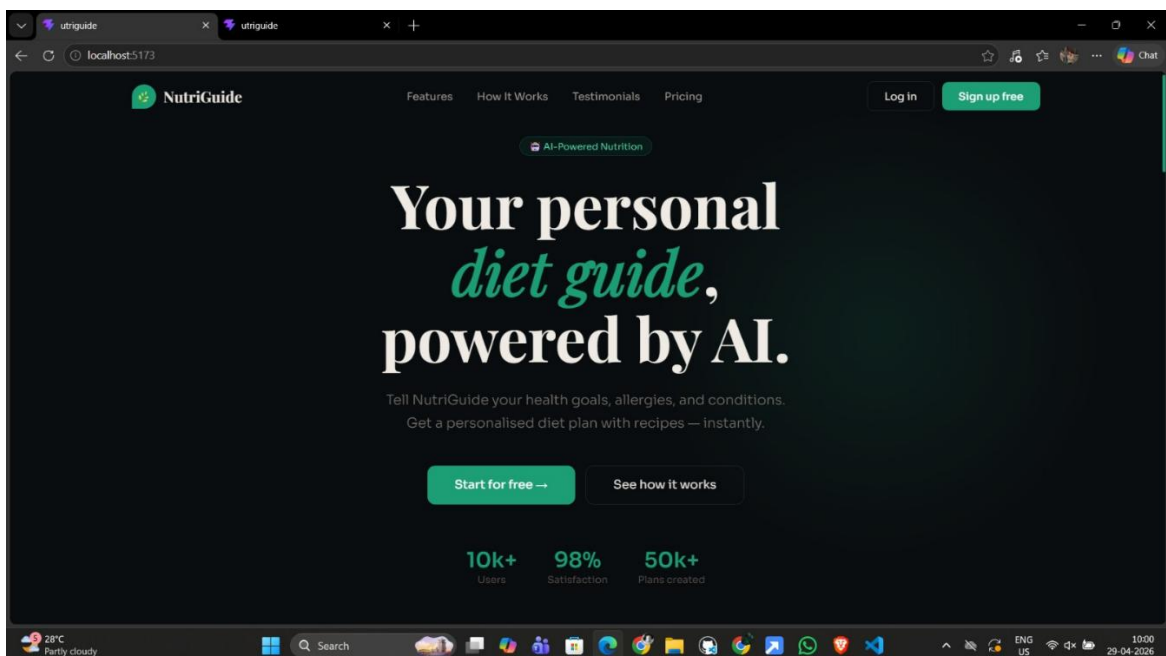
The database architecture of NutriGuide AI is designed with scalability and maintainability in mind. As the platform evolves, additional features such as advanced nutrition analytics, fitness integration, personalized health reports, and AI-based predictive recommendations can be incorporated seamlessly using the existing database structure. The modular nature of the backend system allows developers to expand collections and functionalities without major changes to the existing architecture.

Additionally, efficient database indexing and optimized query handling improve application performance and reduce response time during API communication. MongoDB’s flexible document-based structure enables the application to handle varying nutrition data and user preferences dynamically. This combination of secure authentication, scalable database design, efficient schema management, and AI integration transforms NutriGuide AI from a simple meal tracker into an intelligent and adaptive digital nutrition management platform.

5. Program's Structure Analysing and GUI Constructing

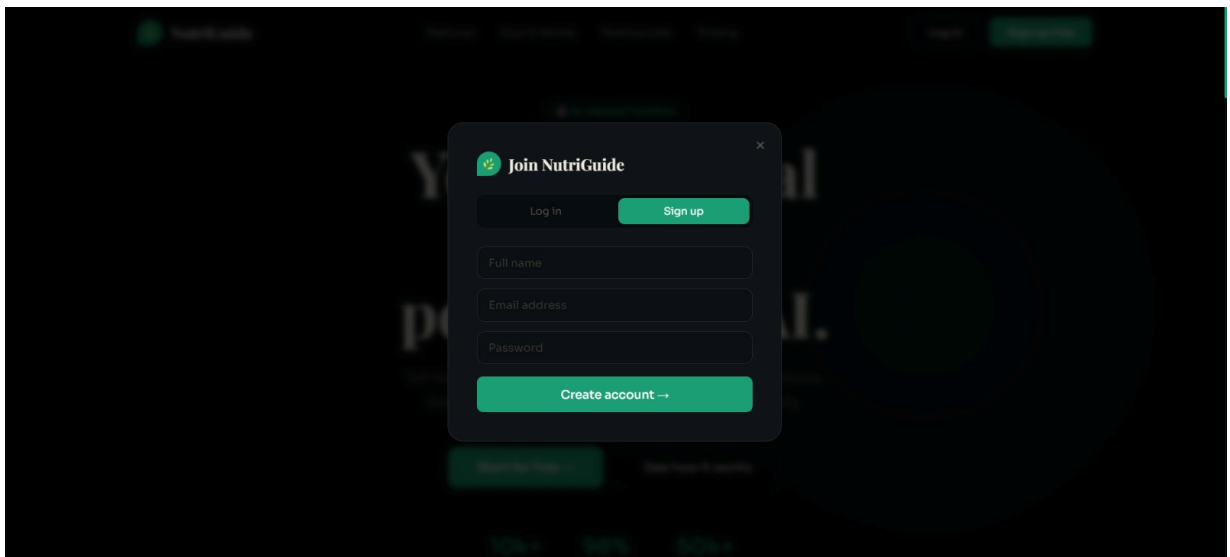
5.1. Home Page

It features a modern, minimal dark-themed interface with a bold headline highlighting AI-powered personalized nutrition guidance. The navigation bar at the top provides quick access to key sections like Features, How It Works, Testimonials, and Pricing, along with Log in and Sign up free buttons for user authentication. Visually, the homepage focuses on elegant typography and a clean layout, supported by subtle glowing background effects that create a premium feel. The central section emphasizes personalized diet planning, allowing users to receive customised nutrition plans instantly based on their health goals, allergies, and conditions. Interactive call-to-action buttons such as “Start for free” and “See how it works” encourage user engagement, while the statistics section showcasing users, satisfaction rate, and plans created builds trust and credibility.



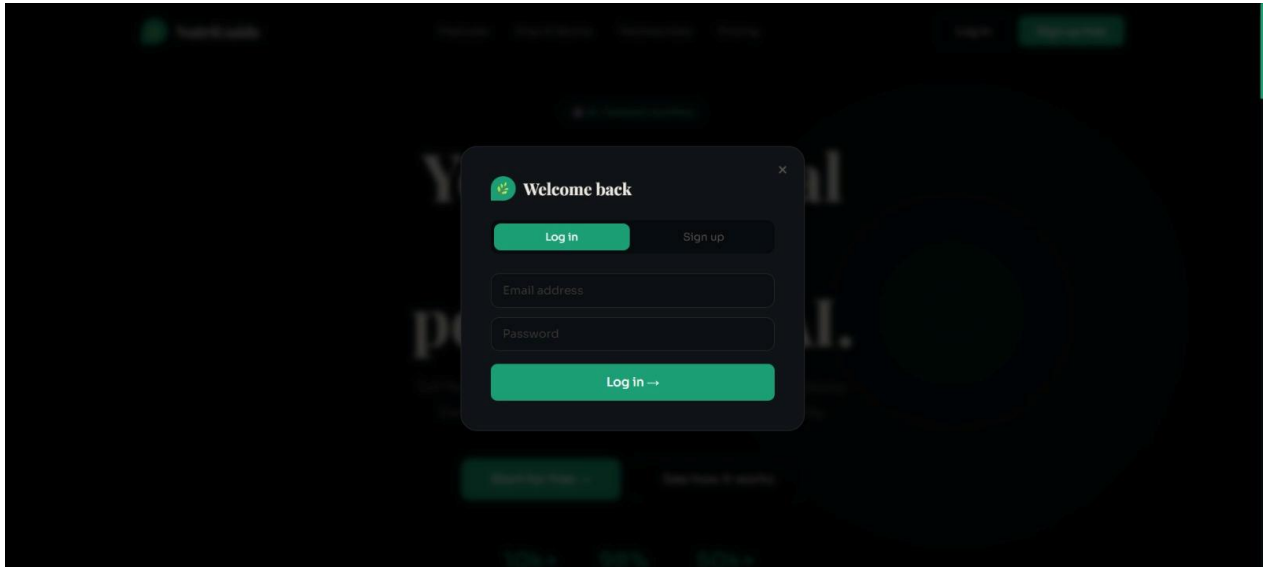
5.2. Sign up Page

This image displays the sign-up modal of the NutriGuide web application. The interface follows a sleek dark-themed design with a modern and minimal layout. At the center of the screen, users are presented with a registration form where they can enter details such as full name, email address, and password to create a new account. The modal also includes toggle options for both Log in and Sign up, allowing existing users to easily switch between authentication modes.



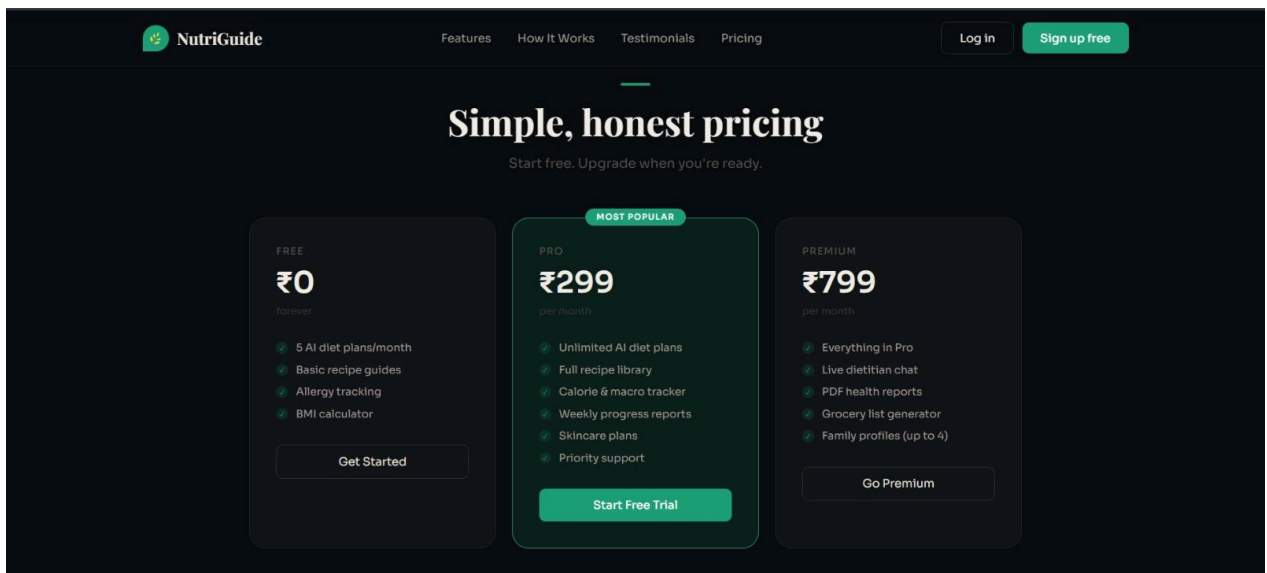
5.3. Login Page

This image shows the login modal of the NutriGuide web application. Users can log in by entering their email address and password. The interface features a modern dark-themed design with options to switch between Log in and Sign up, along with a prominent login button for easy access.



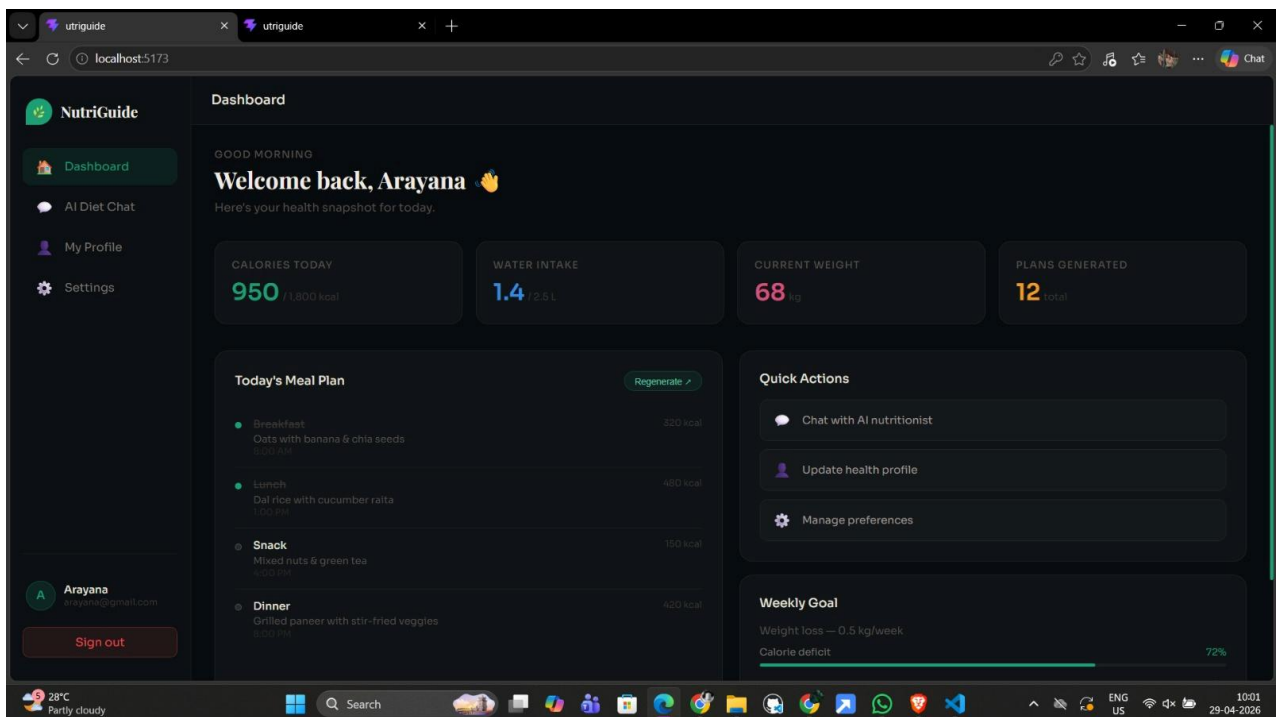
5.4. Subscription Plans

This image shows the subscription plans section of the NutriGuide web application. It presents three pricing tiers — Free, Pro, and Premium — with their respective features and monthly pricing. Users can compare benefits such as AI diet plans, calorie tracking, progress reports, dietitian chat, and family profiles, then choose a suitable plan using the provided action buttons.



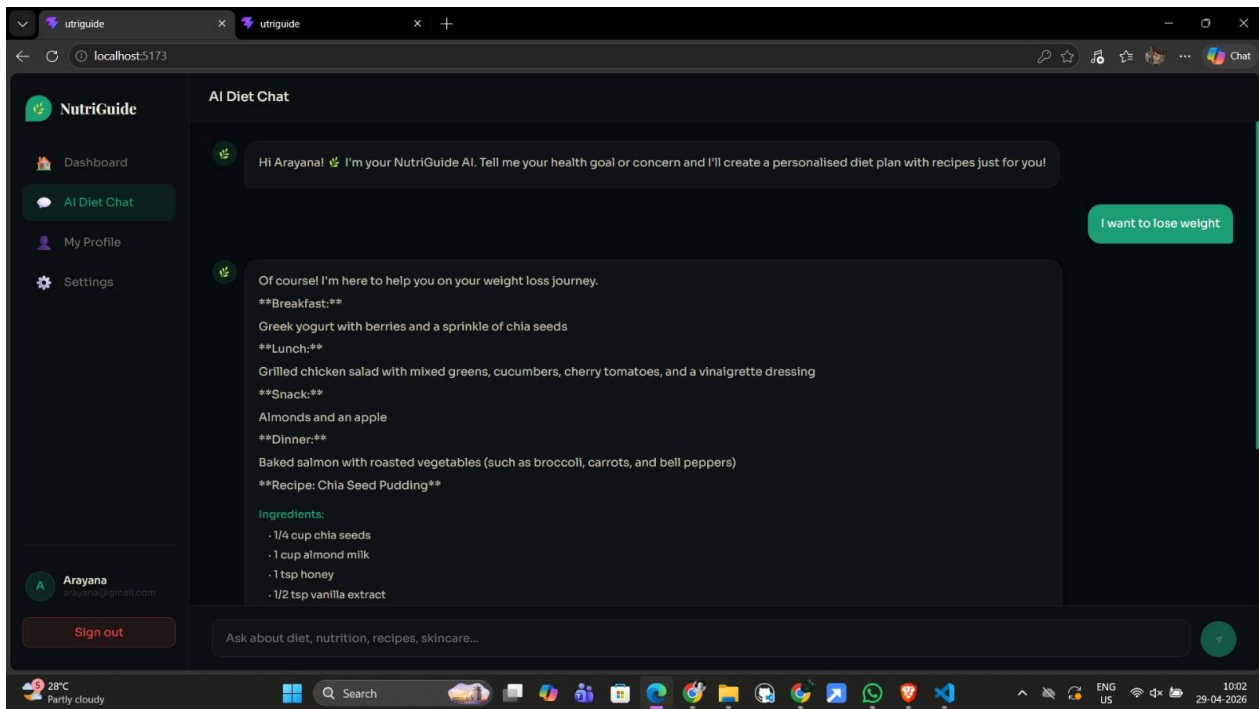
5.5. Dashboard

This image shows the dashboard page of the NutriGuide web application. It provides users with a personalized health overview, including daily calories, water intake, current weight, and total diet plans generated. The dashboard also displays today's meal plan, quick action options like AI diet chat and profile management, and weekly health goals to help users track their nutrition and fitness progress efficiently.



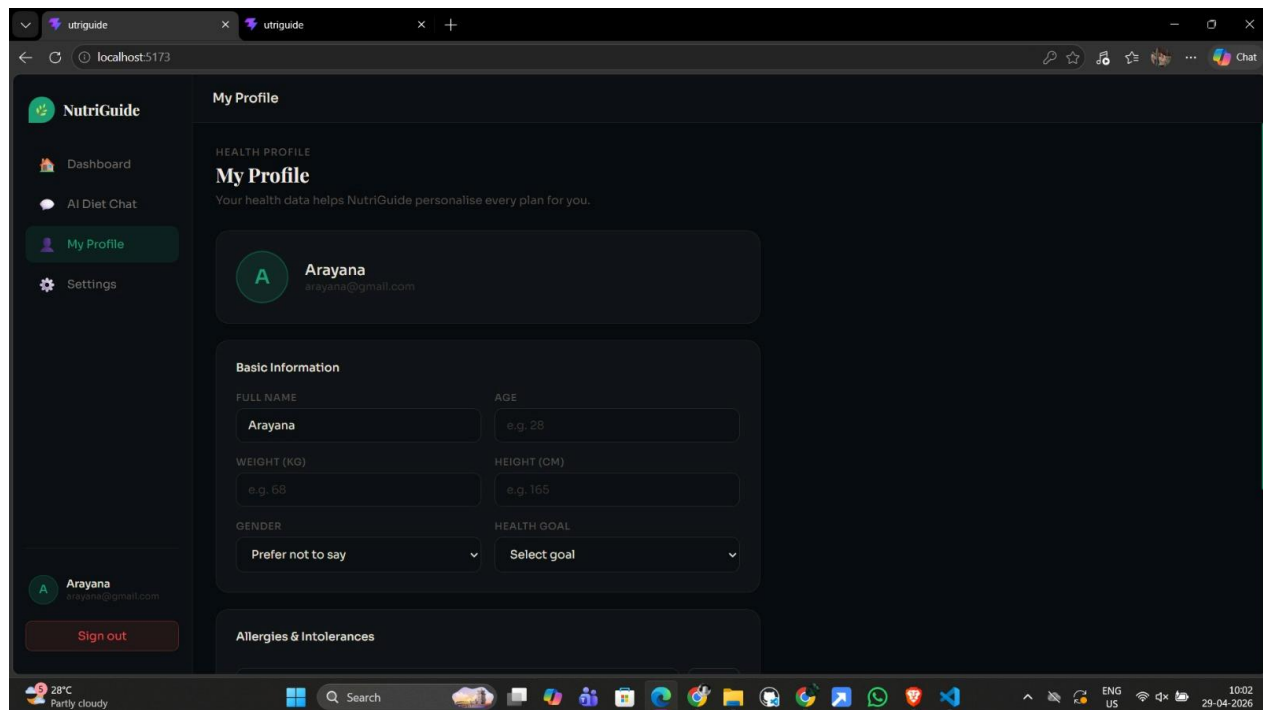
5.6. AI Chatbot

This image showcases the AI Diet Chat feature of the NutriGuide AI platform, where users can interact with an AI-powered nutrition assistant to receive personalized diet plans and health recommendations. Based on the user's health goals, such as weight loss, the system generates customized meal suggestions, recipes, and nutrition guidance in real time. The clean chat-based interface improves user engagement and provides an interactive experience for healthy lifestyle management.



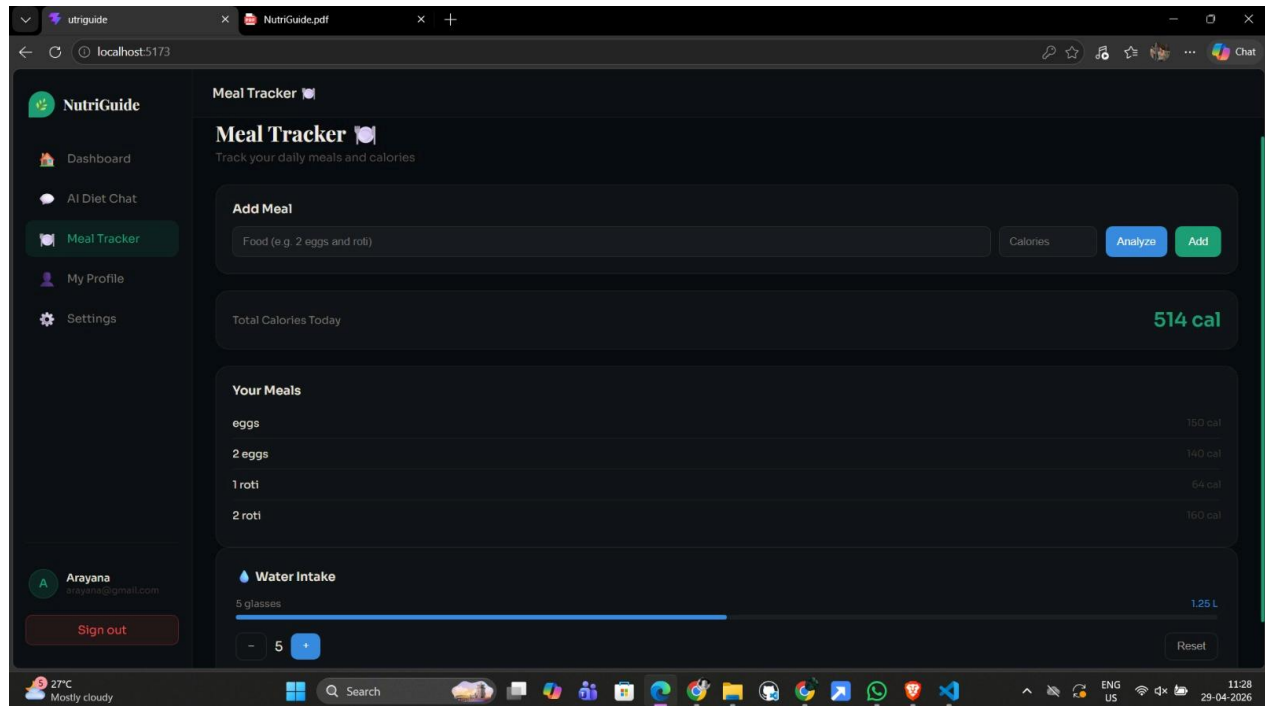
5.7. My Profile

This image shows the My Profile section of the NutriGuide AI platform, where users can manage personal and health-related information such as age, weight, height, gender, and health goals. The page helps personalize nutrition recommendations and diet plans based on user data. The responsive sidebar navigation provides quick access to Dashboard, AI Diet Chat, Profile, and Settings for a smooth user experience.



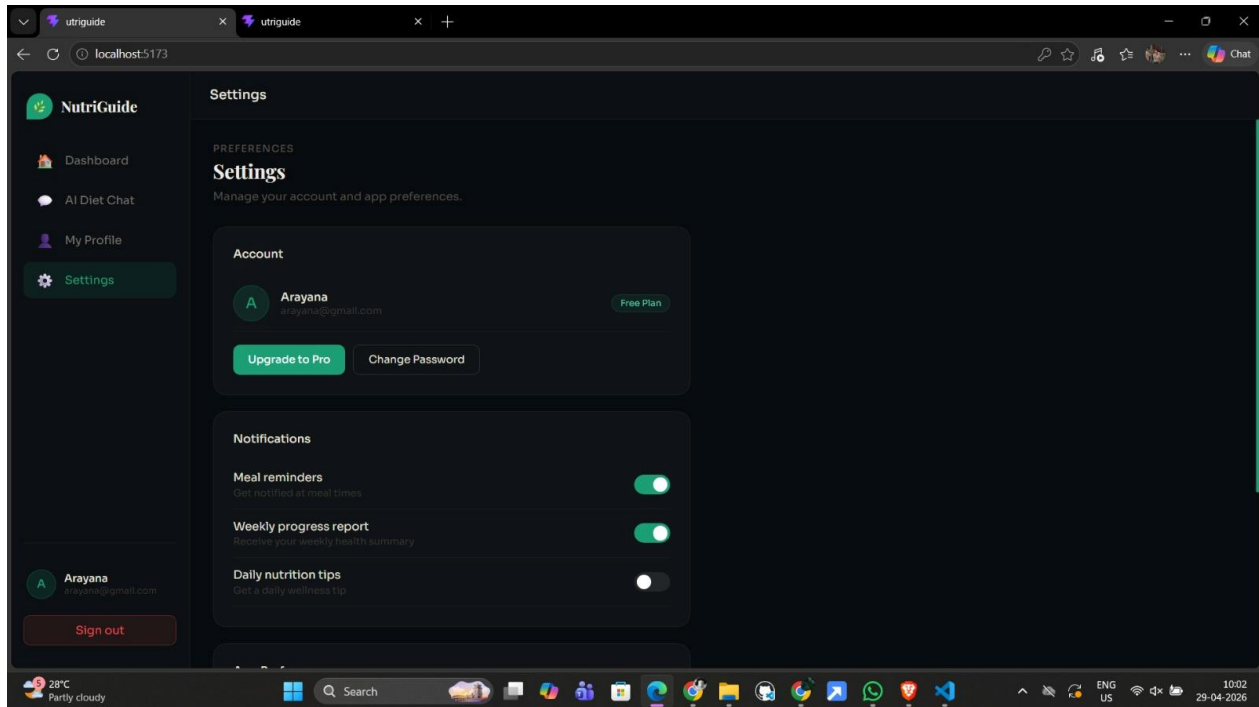
5.8. Meal Tracker

The Meal Tracker provides a streamlined interface for logging daily nutrition. It includes an "Add Meal" field where users can input food items and a dedicated "Your Meals" list that displays specific entries like eggs and roti alongside their calorie counts. The system automatically calculates these inputs to present a prominent "Total Calories Today" summary, helping users monitor their progress toward nutritional goals.

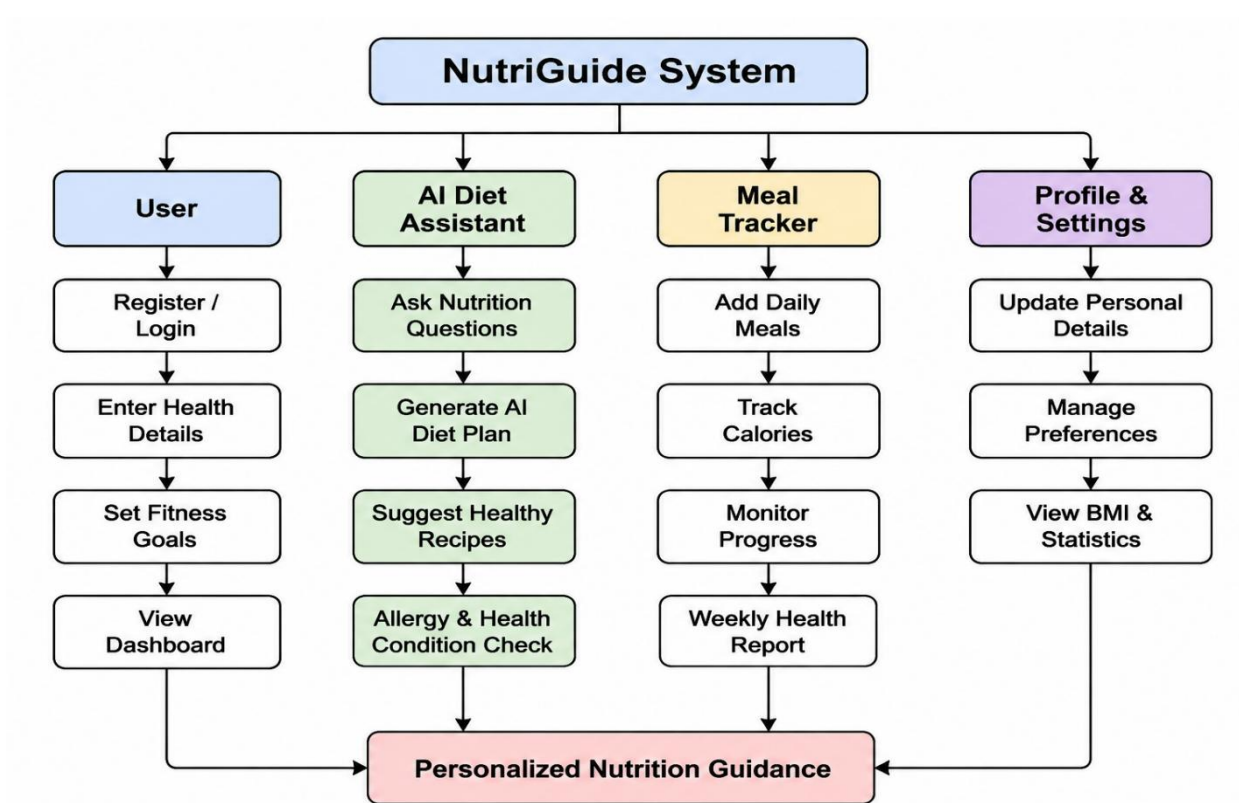


5.9. Settings

The **Settings** interface in this image allows users to manage their account and app preferences through a clean, organized dashboard. The **Account** section displays the user's profile information, current "Free Plan" status, and provides clear options to "Upgrade to Pro" or change their password. Additionally, a **Notifications** area features toggle switches for personalizing alerts, such as meal reminders, weekly progress reports, and daily nutrition tips, giving users full control over their engagement with the platform.



5.10. Flowchart – Representing what we actually do



6. Limitations

While NutriGuide AI provides a modern and intelligent nutrition management platform with AI-powered recommendations and meal tracking functionalities, the system still has certain limitations that affect its overall capabilities and scalability. One of the primary limitations is its dependency on third-party AI services and APIs, particularly the Google Generative AI API used for personalized nutrition guidance and meal recommendations. If the API experiences downtime, usage restrictions, pricing changes, or endpoint modifications, it may directly impact the reliability and responsiveness of the AI assistant. Additionally, API rate limits can affect the system's ability to handle a large number of simultaneous users efficiently during high traffic periods.

Another important limitation is the platform's dependence on a stable internet connection for real-time functionality. Since NutriGuide AI operates as a web-based application with cloud-connected services and AI integration, users require uninterrupted internet access to interact with the system, retrieve nutrition data, and generate AI recommendations. The application currently does not provide full offline support, which may reduce accessibility for users in low-connectivity environments. Although some frontend data can remain temporarily cached, the core functionalities still rely heavily on internet availability.

From a personalization perspective, while the platform generates AI-based nutrition suggestions and tracks user meals, the current version does not include advanced health analytics, fitness tracker integration, or real-time biometric monitoring. Features such as integration with smartwatches, wearable fitness devices, hydration tracking, and personalized medical nutrition analysis are not yet implemented. These limitations restrict the depth of health insights that can be provided to users.

The platform also has limited multilingual and accessibility support. Although the interface is responsive and user-friendly, features such as regional language support, voice-based interaction, screen reader optimization, advanced keyboard navigation, and high-contrast accessibility modes are currently minimal. This may affect usability for users with disabilities or users from diverse linguistic backgrounds. Improving accessibility compliance and

inclusive design remains an important area for future enhancement.

Additionally, NutriGuide AI currently functions primarily as a nutrition management and recommendation system rather than a complete healthcare solution. The AI-generated suggestions are intended for general guidance purposes and should not replace professional medical or dietary consultation. Since the recommendations are based on user-provided inputs and AI-generated responses, there may be occasional inaccuracies or generalized suggestions that do not fully address specific medical conditions or dietary restrictions.

Lastly, although the application is designed using scalable technologies such as React.js, Node.js, and MongoDB, there is still room for optimization in terms of large-scale deployment, performance tuning, and advanced security implementation. Features such as role-based administration, advanced analytics dashboards, cloud-native microservices architecture, and enterprise-level monitoring are currently limited in the present version. Addressing these limitations in future development phases will help transform NutriGuide AI into a more scalable, intelligent, secure, and comprehensive digital nutrition management platform.

7. Conclusion

The development of the NutriGuide AI platform demonstrates how modern web technologies and Artificial Intelligence can transform traditional nutrition management systems into intelligent, interactive, and user-centric digital solutions. By utilizing React.js for building a modular and component-based frontend, Tailwind CSS for responsive and visually appealing design, and Axios for efficient API communication, the application delivers a smooth and engaging user experience across multiple devices. The integration of Node.js and Express.js for backend services, along with MongoDB for scalable database management, provides a strong foundation for secure data handling, reliable performance, and real-time interaction throughout the system.

This project successfully addresses several limitations found in traditional diet and calorie tracking applications by introducing AI-powered nutrition recommendations, personalized meal guidance, calorie monitoring, and secure authentication mechanisms. The integration of Google Generative AI APIs enhances the system's ability to provide intelligent and customized dietary suggestions based on user preferences and nutritional goals. Features such as meal tracking, nutrition dashboards, responsive interfaces, and user profile management improve usability and encourage healthier lifestyle habits among users.

Overall, NutriGuide AI demonstrates the practical implementation of full-stack web development and AI integration in the healthcare and wellness domain. The project not only fulfills the core objective of simplifying nutrition management and meal tracking but also establishes a foundation for future innovations such as advanced health analytics, fitness integration, multilingual support, and intelligent health monitoring systems. By combining responsive design, scalable architecture, and AI-powered personalization, NutriGuide AI represents a modern and forward-thinking solution for digital nutrition and wellness management.

8. Future Scope

In future versions of NutriGuide AI, the platform can be enhanced with advanced AI-driven health and nutrition analysis features to provide a more personalized and intelligent user experience. One major improvement could be the integration of image-based food recognition using computer vision and Artificial Intelligence. By allowing users to upload photos of meals, the system could automatically identify food items, estimate calorie values, analyse nutritional content, and generate accurate diet suggestions. This feature would simplify meal tracking and reduce the need for manual food entry, making the application more efficient and user-friendly.

Another important future enhancement is the integration of wearable fitness devices and health tracking systems. NutriGuide AI can be connected with smartwatches and fitness trackers to collect real-time health data such as step count, heart rate, sleep patterns, hydration levels, and calorie expenditure. By analysing this information alongside user meal records and dietary preferences, the platform could generate highly personalized nutrition plans and fitness recommendations tailored to individual lifestyles and health conditions.

The application can also be expanded to include advanced health analytics and predictive insights powered by Machine Learning algorithms. By analysing user behaviour, nutrition history, and fitness data over time, the system could predict unhealthy eating patterns, identify nutritional deficiencies, and recommend preventive health measures. AI-driven progress tracking and personalized wellness reports could further improve user engagement and help individuals maintain long-term healthy habits.

Additionally, NutriGuide AI can be enhanced with multilingual support and voice-enabled interaction to improve accessibility and usability for a larger audience. Voice-based AI assistants could allow users to interact with the system through natural language conversations for meal recommendations, calorie tracking, and nutrition guidance. Regional language support would make the platform more inclusive and accessible to users from diverse linguistic backgrounds.

Future versions of the platform may also include social and community-based features where users can share diet plans, participate in fitness challenges, and interact with certified nutritionists or dietitians through online consultations. This would transform NutriGuide AI from a standalone nutrition tracker into a comprehensive digital wellness ecosystem that encourages healthier lifestyles through collaboration and expert guidance.

Furthermore, cloud-based scalability and mobile application support can significantly improve the reach and performance of the platform. Dedicated Android and iOS applications with real-time notifications, reminders for meals and hydration, and offline data synchronization can enhance user convenience and engagement. Integration with cloud analytics and AI optimization systems would allow the application to handle larger user bases efficiently while continuously improving recommendation accuracy and overall system performance.

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